

CECS 451 · Phase 4 · Term Project

SocialAgent

Say it. Schedule it. Show up.

A 3-container Discord bot powered by a local Gemma LLM that negotiates group events and auto-schedules them.

Nicholas · Alfredo · John



Coordination Friction Kills Plans



Too Many Apps

Juggling Partiful, Google Forms, Doodle, and group chats creates decision fatigue.



Already Downloaded discord

No need to download any other apps to manage just upload the bot to a server give it permission to read the thread and done.



Static Scheduling

Traditional forms can't adapt. Once sent, a Partiful invite is a one-way street.

Market Gap: No existing tool combines natural language, fair location math, and automatic scheduling in one Discord-native experience.

SocialAgent

negotiator



1. Natural Language

User types event request in Discord chat



2. Gemma LLM + OSM

Local AI parses intent, calculates best time block



3. Auto-Schedule

Discord API V10 event created instantly

NLP Parser

Gemma/Ollama

OpenStreetMap

Discord API V10

Docker × 3

Core Tech Stack Synergy

The Brain



parser.py

NLP Parser

Converts messy natural language into machine-readable timestamps. Acts as a robust fallback so the AI always has structured context — even when user input is ambiguous.

Spacy

Regex chains

Fallback logic



location_service.py

Venue Scout

OpenStreetMap

Overpass API

Centroid math



scheduler.py

Agentic Heart

Takes the parsed timestamp and centroid location to perform weighted conflict resolution. This is the 'agent' — it decides the final event time and venue autonomously.

Conflict resolution

Weighted scoring

Gemma prompts



database.py

Memory Layer

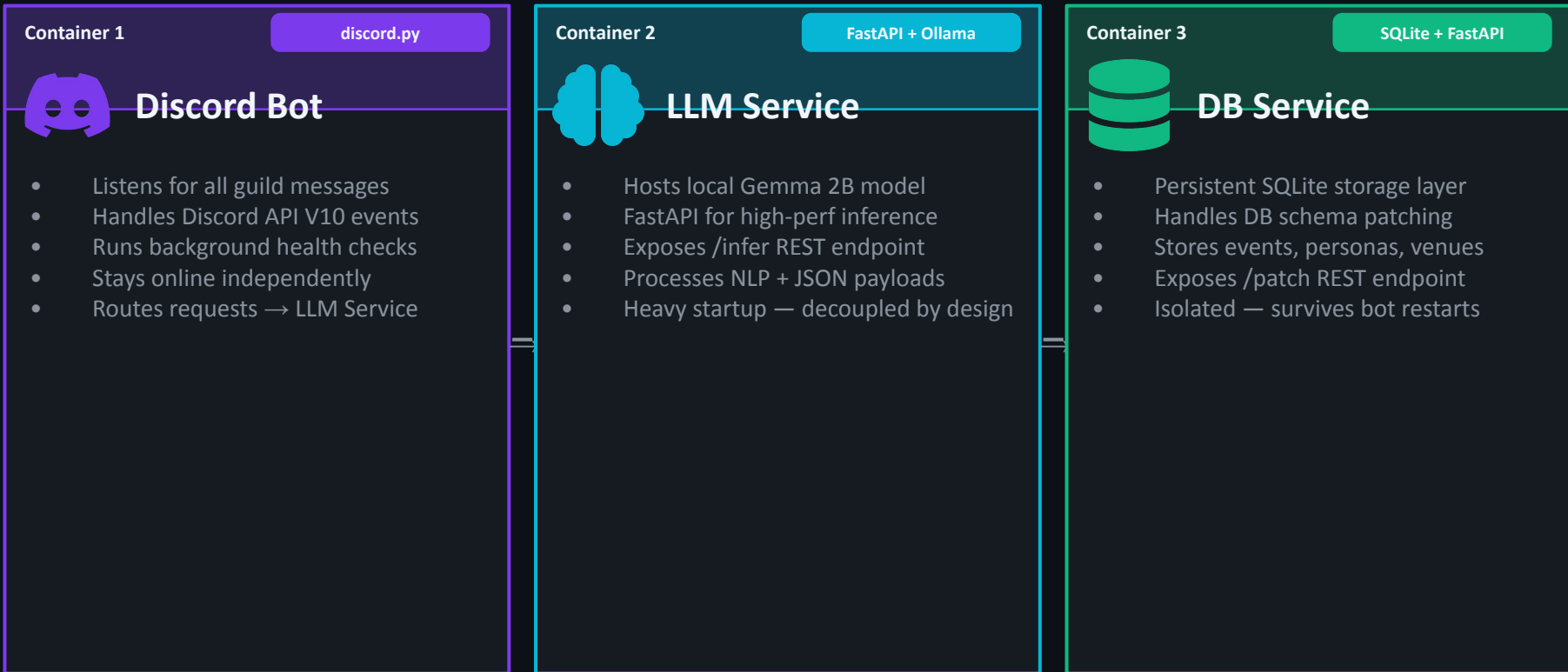
Ties the whole stack together. Stores user personas, venue history, and event records. Every module reads/writes here — it's the shared brain state.

SQLite

User personas

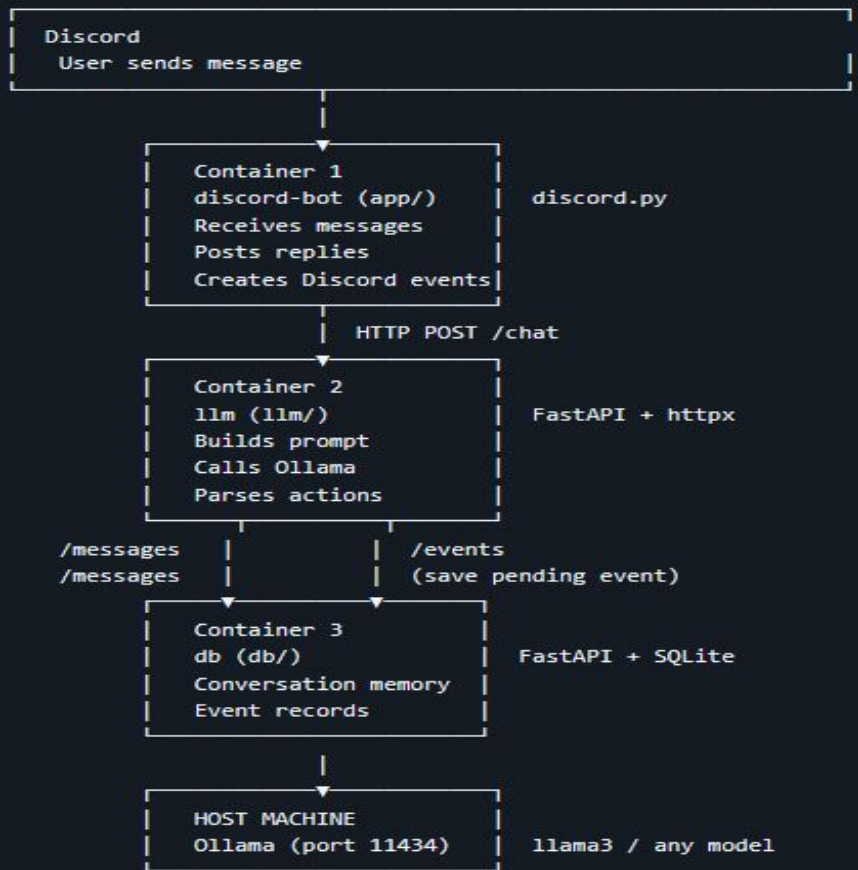
Venue history

3-Container Microservice Architecture



Decoupled design → each container restarts independently with zero cascading failures across the pipeline.

A 3-container event coordination assistant powered by Discord + Ollama.



Project Structure

```
event-planner/
├── docker-compose.yml
├── .env.example
├── app/                # Container 1 – Discord bot
│   ├── Dockerfile
│   ├── requirements.txt
│   └── bot.py
├── llm/                # Container 2 – LLM orchestration
│   ├── Dockerfile
│   ├── requirements.txt
│   └── app.py
└── db/                # Container 3 – SQLite memory
    ├── Dockerfile
    ├── requirements.txt
    └── app.py
```

DEMO

Natural language event creation · Centroid calculation · Discord calendar auto-schedule



Wawa Yesterday at 10:53 PM

yeah it be to slow anyway

@Eve so @noj @johongir and I are gonna go get KBBq on Saturday then maybe get like a drink afterwards. Probably boba. Can you @ those two so we can plan this. Im looking at like 6:00PM



Eve **APP** Yesterday at 11:06 PM

Eve: Sounds good — Saturday at **6:00 PM** works as a starting point. I'll mention @noj and @johongir so y'all can lock in the KBBQ → boba plan. If anyone wants to tweak the time, just say so.



Wawa Yesterday at 11:06 PM

all looks good just schedule it



Eve **APP** Yesterday at 11:06 PM

Eve: Got it — I'm scheduling **KBBQ + boba** for **Saturday at 6:00 PM**. See you all there!



Event scheduled! [@Wawa](#)

KBBQ + boba — Saturday, May 16, 2026 11:00 AM to 1:00 PM

<https://discord.com/events/1503600654055116991/1503639608024498326>

Where SocialAgent Lives in Society

USE CASES



HR & Onboarding

Auto-schedule team lunches, offsites & interviews across timezones. No more 30-email chains. No more guessing on whos is going to be there



Company-Wide Planning

Town halls, all-hands, and cross-department events planned in one Discord command.



University Clubs

Student orgs, Greek life, and sports teams coordinate meetings without leaving Discord.



Healthcare Shift Teams

Nurses and shift workers find fair meetup windows based on rotating schedules.



Event Promoters (live nation, club events, socials)

Concert & festival promoters use the centroid engine to pick venues that serve the most fans.

MONETIZATION \$\$\$



Sponsored Venue Ads

Businesses pay to appear as "Featured Venue" suggestions when the centroid algorithm picks nearby locations. Think Google Maps ads but inside Discord.



B2B SaaS Licensing

Sell white-labeled SocialAgent to HR platforms (BambooHR, Lattice) and event tools (Eventbrite, Cvent) as an embedded scheduling API.



Enterprise Analytics Tier

Charge companies for anonymized attendance trend reports, peak planning windows, and team engagement scores derived from event data.



Premium Bot Subscription

Free tier: 5 events/month. Pro tier (\$9.99/mo): unlimited events, calendar sync, dietary preference memory, and priority Gemma inference.



Large-Scale Event Planner API

Festivals, corporate retreats, and wedding planners pay per-API-call for centroid routing, vendor pricing scrapes, and group availability windows.

Competitive Matrix

Feature	Partiful	Standard Bots	SocialAgent ✓
Natural Language	✗	Partial	✓ Full NLP
Auto Scheduling	Manual	✗	✓ Discord API V10
Real time planner	✗	limited	✓ Discord thread
API Cost	SaaS Fees	Paid Tiers	✓ \$0 Lean Stack
Data Privacy	Cloud	Cloud	✓ 100% Local RAG

3 Business Pillars



\$0 Operating Cost

Lean Stack Architecture

- Gemma 2B runs locally via Ollama
- OpenStreetMap — zero licensing fees
- Self-hosted on any Docker machine
- vs. GPT-4 API: ~\$0.03/1K tokens saved



100% Data Sovereignty

Local Vector RAG

- User data never leaves the server
- Local embeddings — no cloud calls
- GDPR/CCPA ready by design
- Competitive privacy moat vs SaaS

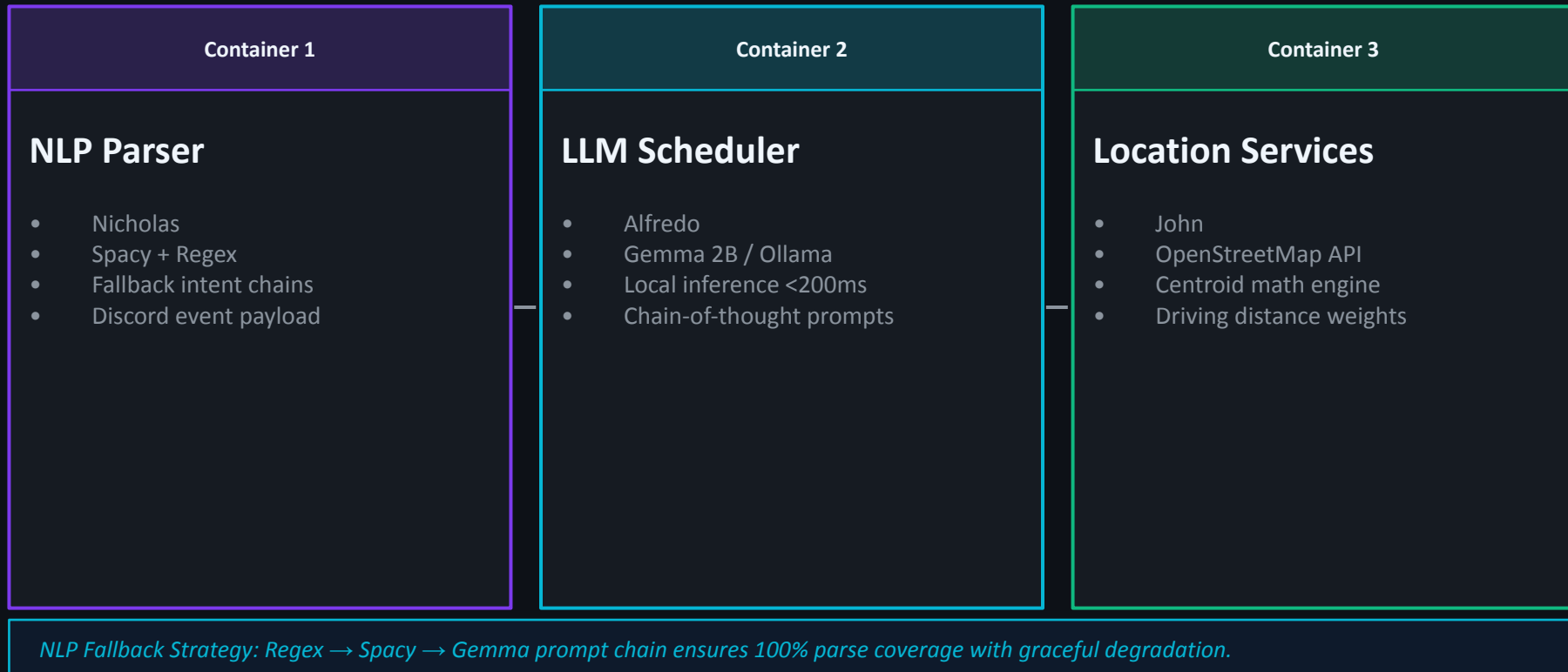


Spatial Fairness Engine

Centroid Optimization

- Geographic centroid of all members
- OSM walking/driving distance weights
- Eliminates location bias disputes
- Provably fair — algorithm is auditable

3-Container Docker Architecture



Integration Tests & AI Metrics

100%

test_spatial_math.py

100%

test_action_payload.py

<200ms

Gemma Local Inference

\$0

External API Cost

Inference Latency Comparison (ms)



CI/CD Pipeline: All tests run on every push. 100% pass rate maintained across 3 containers with isolated pytest suites.

Where SocialAgent Goes Next

Phase 5 — Near Term

- Passive voice-channel monitoring for availability signals
- Web scraping for real-time venue pricing & capacity data
- Instagram Reels transcript parsing for event inspiration

Phase 6 — Mid Term

- Calendar integration (Google/Outlook) for true availability sync
- Dietary & accessibility preference database per user
- Spatial engine

Phase 7 — Long Term

- Mobile companion app with push notifications
- Venue partnership API for exclusive group discounts
- Fine-tuned SocialAgent-7B model on event negotiation data

Microservice Contributions



Nicholas

NLP Engineer

Container 1 — NLP Parser

- Spacy pipeline & Regex fallback chains
- Discord event payload schema design
- Intent classification & entity extraction
- CI tests: test_action_payload.py



Alfredo

AI / LLM Engineer

Container 2 — LLM Scheduler

- Gemma 2B integration via Ollama
- Chain-of-thought prompt engineering
- Discord API V10 event scheduling
- Local inference optimization



John

Location & OSM Engineer

Container 3 — Location Services

- OpenStreetMap tile & routing API
- Geographic centroid algorithm
- Driving distance weight calculations
- CI tests: test_spatial_math.py

SocialAgent

Say it. Schedule it. Show up.

\$0 Cost

100% Private

ZERO EFFORT

- ✓ Performance Evaluation — CI Tests + Latency Benchmarks (10 pts)
- ✓ Business Impact — Competitive Matrix + Cost + Privacy (10 pts)
- ✓ Presentation — Prototype Demo + Team Roles slide (5 pts)